

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (CE) Sem - I University Examination, 2010-11

CE702:Advanced Computer Network

Date: 04.01.2011, Tuesday Time: 10:30 a.m. To 01:30 p.m. Maximum Marks: 70

- Instructions:**
- (1) All questions are compulsory.
 - (2) Indicate clearly, the option you attempt along with its respective question number.
 - (3) Figures to right indicate marks
 - (4) Rough work is done in the last page of main supplementary, don't write anything on the question paper.

SECTION-I

Q-1 Answer the following questions. [11]

- [A] How does RIPv2 packet format differ from RIPv1 packet format? [2]
[B] A host with IP address 137.23.232.232/19 sends a packet to a host with IP address 137.23.216.216/19. Is it direct delivery or indirect delivery? [2]
[C] What is proxy ARP? What are the advantages and disadvantages of it? [2]
[D] What is anycast address? Which protocol supports anycast address at network layer? [2]
[E] Draw and explain the CSMA/CA flow diagram. [3]

Q-2 Answer the following questions. [12]

- [A] What is use of strict-route and loose-route option in IPv4 address? [4]
[B] What is packet format of ATM ARP? What are the ATM ARP Operations? [8]

OR

- [B] By taking suitable network topology, explain how does OSPF make routing table. [8]
(Note: Explain process for only one router)

Q-3 Answer the following questions.(Attempt any three) [12]

- [A] Which protocol is example of Border gateway protocol? Explain how does it work?
[B] In which possible cases/scenario, ARP is used.
[C] What are the types of link used in OSPF?
[D] What are the components and modules of IP packet processing? Draw architecture of it.

SECTION-II

Q-4 Answer the following questions. [11]

- [A] There are three types of socket interface. Stream socket, datagram socket and Raw socket. What is use of each? [2]
[B] Is it possible to have connection-oriented iterative server? In any case, write down advantages or disadvantages of it. [2]
[C] How DNS can be attacked? How can it be protected? [2]
[D] What is "half-close" case in TCP? [2]
[E] Write the comparison between IPv4 and IPv6 headers. [3]

Q-5 Answer the following questions.

[A] By taking suitable example, write down Byte Ordering Functions used in socket programming. [12]
[4]

[B] What are the types of Resource Records in DNS? What is the use of each? What is zone, domain, root server and primary server with respect to DNS? [8]

OR

[B] How is silly window syndrome arisen? What are the solutions of it? [8]

Q-6 Answer the following questions.(Attempt any three)

[A] What is autoconfiguration in IPv6? What are the steps performed by computer to assign autoconfigured IPv6 address? [12]

[B] Explain CBT Multicast Routing Protocol.

[C] What are the congestion policies in TCP to deal congestion?

[D] Draw TCP header format. Explain fields of it.